Who can challenge, correct and decide?

A close reading of Magnifica Humanitas and its formal governance findings

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1. The encyclical and this study

What we have done. We have translated selected AI-governance commitments in Pope Leo XIV's Magnifica Humanitas into explicit requirements and checked examples using Lean, a computer program that verifies logical proofs. We built hypothetical arrangements where a protection fails, then checked repairs that supply it. For example, an appeal can be reviewed while a benefits denial remains unchanged because the finding never reaches an applied correction. Our proposal is to use these source-linked comparisons to examine whether an institution's procedures deliver its stated protections. This report explains the encyclical, the requirements we selected, and what the checked comparisons show.

Why it matters. Imagine receiving a benefits denial with a named agency, a help desk and an appeal form. Those provisions still leave questions: who is assigned to your decision, can anyone correct it, and does a successful review actually change the record? Our work makes these connections explicit. The encyclical places technological change within a Christian account of human dignity and the common good (§§1-16); its opening contrasts Babel's self-aggrandizement with Nehemiah's cooperative rebuilding (§§7-10). We examine what selected commitments require in a decision procedure. The contribution is a way to distinguish a stated protection, the powers available to provide it, and the performed action that makes it effective.

The encyclical’s wider argument. Magnifica Humanitas is an encyclical, a papal teaching letter, dated 15 May 2026 and addressed to Catholics, other Christians and people of goodwill (§16). Its official English PDF has 91 pages, including notes, and 245 numbered paragraphs. Its five chapters move from social doctrine's development and foundations to technology and dominance, truth/work/freedom, and power/peace; the conclusion returns to Christian life and hope. This wider argument gives our investigation its setting. An appeal desk is one practical arrangement through which a commitment might operate; it cannot represent the whole ethical vision. Our study therefore identifies a particular group of passages and makes its interpretation of them available for inspection.

Our proposal and source selection. We select §§97-111, on pages 33-38 in Chapter Three, because their responsibility, recourse, oversight, participation and design questions can be examined together. Picture following a benefits denial from its calculation to a named officer, an appeal, a review and an applied correction. Our proposal is to examine that chain against explicit, source-linked requirements and use checked repairs to identify missing connections. The following page explains our close reading; pages 3-5 develop the formal findings through examples in employment, public benefits and high-stakes autonomy. Source principles, our procedural choices and the proof results remain distinguishable throughout, so readers can inspect both the interpretation and the evidence.

Source: Leo XIV, Magnifica Humanitas, official English edition, 15 May 2026; introduction, chapter structure and selected §§97-111. Page numbers refer to this edition, accessed 2 November 2026. The full document is contextual background; the formal source boundary remains §§97-111.

2. A close reading of the selected passages

What we selected, how we read it, and why the questions belong together

From context to consequential decisions. Paragraphs 97-103, on pages 33-36, supply context about human direction, AI use and consequential decisions. Consider an applicant told that a computer placed her below an eligibility threshold. She needs a route to question that particular outcome and someone assigned to answer for it. This is why we focus on decisions affecting identifiable people: the relationship between a protection and its intended recipient can be examined precisely. We use the surrounding observations to understand the setting, rather than treating every empirical description of AI as a permanent rule. Employment and public services supply concrete settings for asking how responsibility reaches the person affected.

Appeal and accountable responsibility. On page 36, §104 addresses exclusion “without the possibility of appeal”; §105 requires that “responsibility must be clearly defined at every stage”. We read these commitments together as grounds for examining a connected chain. Suppose a denied applicant is directed from a help desk to a review team, but neither identifies the person assigned to the denial. Several departments are involved, yet the applicant still lacks that answer. An accessible appeal, an identifiable assignee and correction power are distinct pieces of evidence in our interpretation. The question is whether a proposed procedure joins them for the same case, rather than leaving each protection somewhere in the organization.

Institutions and participation. Paragraphs 106-109, on pages 36-37, extend the discussion to institutional safeguards, participation and correction. Suppose residents submit a request to pause a disputed benefits decision. Recording that request shows that input was received; the procedure must establish separately whether the request can cause a pause. Similarly, a board authorized to review routine inquiries may lack authority over this denial. We read the selected passages as prompting examination of such practical powers and their scope. Our models compare receipt, influence, binding effect and intervention without making one mechanism a complete definition of participation or subsidiarity. Wider questions, including the distribution of data and resources, remain outside these comparisons.

Authority, development and our interpretation. On page 37, §110 challenges the assumption that “technical power automatically confers the right to govern”; §111, continuing onto page 38, addresses developers. An operator might be able to make a review rule produce a grant or a denial while lacking institutional recognition to govern that decision. A developer might be assigned to design review while a separate official handles the later denial. These examples motivate our separate questions about capability, mandate and stage-specific responsibility. The resulting analytical categories are ours, and the particular board or correction desk is a proposed implementation. Keeping that distinction visible allows readers to examine how our procedural interpretation relates to the cited text.

Source locations: §§97-103, pp. 33-36; §§104-107, p. 36; §108, pp. 36-37; §§109-110, p. 37; §111, pp. 37-38. Quotation marks identify exact short extracts; other source descriptions are attributed paraphrases. The procedural questions and interpretations are the study’s analysis.

3. Responsibility and recourse

What the formal comparisons establish about a particular decision

How the checking works. Suppose an office promises appeal but accepts no appeal from the person denied. We describe that arrangement precisely and ask Lean to check whether it meets our stated appeal requirement. We then describe an accessible channel and check the repair. This separates a duty from its fulfillment and supplies inspectable comparisons of failure and success. Lean verifies that the conclusions follow from the written rules; the source interpretation and hypothetical facts are stated inputs. The following examples explain those rules in ordinary language. Their scores, roles and procedures are modeling choices, not measurements of actual agencies.

Find the person assigned to your decision. For §105 (p. 36), our benefits example distinguishes a public contact who handles inquiries from the officer assigned to a denial. A separate notice comparison gives a truthful score of 47, a cutoff of 50, published safeguards and an identifiable contact. The actual denial officer can still be unidentified. Lean checks this gap and a repair that names the already-assigned officer while preserving the assignment and channels. It also checks the sufficient connection from responsible institution through case allocation to identifiable assignee. Being named does not itself supply correction power. For the applicant, the concrete questions are who is assigned to this denial and what that person can do about it.

Follow responsibility through the handoff. For §§103, 105 and 111 (pp. 35-38), imagine an employment office changing to direct AI production of a decision. Our comparison retains the named officer's assignment, appeal and correction capacity. Lean checks that the tested requirements remain satisfied. Removing the officer's assignment makes all three safeguards policies fail, even while the other channels remain. In a separate design/use comparison, a developer assigned to design review does not thereby fill an unassigned role for a later benefits denial. Assigning a deployer to that denial covers the second stage; sharing the downstream assignment is also compatible. These results concern the specified allocation, not sole legal liability.

Separate review, appeal and correction. For §§104-105 (p. 36), a hiring office can review a rejection internally while accepting no appeal from the applicant. Lean checks that arrangement. Both tested appeal readings reject exclusion without accessible appeal; a broader reading also requires access before exclusion for affected people in high-impact cases. In the employment comparisons, a rejected, harmed applicant can reach a reviewing desk that lacks correction power. Before exclusion, another affected applicant lacks appeal while no harm triggers the separate correction duty. These cases show why wider appeal and harm-triggered correction capacity cannot substitute for each other. An institution must establish access and relevant power under the conditions each requirement specifies.

Make the reviewed appeal change the record. For §§105-106 and 108 (pp. 36-37), consider an assisted-intake benefits denial already stored in the agency's records. Reassessment under the corrected scoring rule yields a grant. A board can complete review while the denial remains unchanged if its finding never reaches the correcting office. Lean checks that the completed path needs open admission, a submitted request, completed review, a working handoff, correction power and an update applied through the correct office. Missing the handoff or leaving the update unperformed preserves the denial; applying it before review also fails. With the modeled checks enforced, the ordered path changes the stored award to grant. Payment remains a separate action.

Evidence: technical paper §§2-5 (method, appeal, assignee, disclosure), §8 (stages) and §11 (applied correction); declaration references and assumptions accompany those sections.

4. Institutions, participation and design

Which powers exist, who holds them, and whether they reach the operative rule

Give the board power over this denial. For §§106 and 108 (pp. 36-37), suppose an agency announces an independent review board, but its remit covers routine inquiries rather than the disputed denial. Our first comparison has that gap. A second board covers the denial but lacks intervention power; a third supplies both. Lean checks the failures and repairs while the person's other safeguards stay fixed. Independence is assumed in the example. In the separate appeal-to-update workflow, the board has a usable intervention path when appeal access, correction power and the handoff exist. The practical test is to follow this denial: may the board review it, and can its finding reach an action that changes the record?

Show what a community request can do. For §§108-109 (pp. 36-37), residents ask to pause the disputed denial before implementation. In our initial pause procedure, their request is recorded but has no effect, while the operator and independent board can each impose a hold. Enabling the local input makes the residents' request effective even when both external holds are released; both external powers remain available. Lean checks that comparison, assuming requests arrive and holds are honored. A separate consultation model shows recommendations influencing outcomes while the institution retains the final choice; a binding rule changes that arrangement. These are specific powers. A temporary pause does not itself award the benefit or correct the record.

Examine the rule behind the explanation. For §§104 and 109 (pp. 36-37), two benefits applicants each have verified eligibility score 60 and cutoff 50. One applies online; the other uses assisted intake. The initial rule deducts 15 points for assisted intake, so that applicant receives a denial while the online applicant receives a grant. Both letters accurately explain the arithmetic. Our adopted case policy says intake channel should not change entitlement. Lean checks that the penalty fails the adopted equal-treatment and common-cutoff tests despite the existing safeguards passing. Removing it supplies both tests. The lesson concerns this explicit criterion: an accurate explanation can reveal a rule that fails the standard chosen to assess it.

Make the objection change the calculation. For §§104-105, 109 and 111 (pp. 36-38), an objection asks to remove that 15-point penalty. Logging it leaves the calculator unchanged. A design reviewer without criteria-changing power cannot implement it; the denial handler's case assignment also supplies no design authority. An assigned, empowered reviewer can implement a proposal that passes the eligibility check. The scorer then uses the returned rule, and both applications yield grants. Lean also checks that any number of permitted revision attempts leaves the corrected rule in place: a proposal to restore the penalty fails validation. That guarantee assumes enforced checks, fixed permissions and the two modeled candidate rules. Earlier denials still need the separate reopening-and-update path.

Separate the ability to act from authority. For §110 (p. 37), an operator in our example can make the review rule choose a grant or a denial. That capability can coexist with no recognized mandate for the decision. Granting the mandate changes recognition while preserving the operator's technical powers. Lean checks these comparisons and a general result for authority interpretations that explicitly require recognition and decision remit. Within that family, capability alone cannot supply authority. The mandate is institutional evidence, not a complete definition of moral legitimacy. The concrete inquiry is who authorized this operator to govern this decision, alongside what the operator can technically do.

Evidence: companion technical paper §§6-10: oversight, participation/local pause, authority, adopted fairness and operative rule correction. The source motivates the questions; the particular procedures and enforcement assumptions are ours.

5. Timing, interpretation and practical use

What the results establish together, and how their evidence can be inspected

See why a late stop is different. For the preventive-control question motivated by §§97, 105-106 (pp. 33-36), picture automated equipment about to cause damage. This illustrates our hypothetical high-stakes model. In one sequence, the stop command reaches the machine before the hazardous action, preventing damage. In the other, damage occurs before the stop arrives. Both records eventually show that the stop arrived and the administrative record was corrected. Their physical outcomes differ. Under our rules, later reviews and record corrections cannot undo the damage. Lean checks that result for any number of the permitted later steps. It also checks that a timely delivered stop keeps a loss-free state protected through those steps. The model assumes delivered stopping is obeyed.

Explain which protections work together. For the tested requirements motivated by §§104-108 (pp. 36-37), consider the same unidentified denial officer under different appeal rules. Opening appeal earlier does not identify that officer. Lean checks that missing assignee identification fails all three safeguards policies; board-coverage and power gaps also persist when those policies pass. Conversely, giving an applicant appeal does not supply a correcting desk, and providing correction after harm need not supply appeal before exclusion. The additions require separate evidence. A repaired benefits case meets eight tested combinations of responsibility, appeal and oversight requirements, with correction capacity retained. This shows those protections can work together; it does not establish every conclusion under every possible reading of the encyclical.

Use the checked comparisons to examine a procedure. We propose using these source-linked, checked comparisons as reference cases when examining procedures intended to implement the selected commitments. An agency could take one denial notice and trace it to the actual assignee, the accepted appeal, the authorized reviewer and the office that applies the correction. If the reviewed finding stops between departments, the procedure has a specific connection to repair. If the objection concerns the scoring rule, the next question is who can revise that rule and how an approved revision reaches the calculator. This method gives institutions concrete grounds for improving procedures and affected people concrete questions to ask. It makes implementation evidence visible alongside the ethical requirement being applied.

What the work contributes. We have made selected commitments in the encyclical precise enough to examine through machine-checked failures, working repairs and conditional guarantees. The completed models show why an appeal form, a named office and a review board can coexist with an unchanged denial, and how a connected review-to-update path corrects the stored award. The contribution is the traceable relationship between the actual passages, our explicit interpretations and the checked consequences. The technical paper and reproducible Lean files let readers inspect that evidence and its assumptions. Institutions applying these commitments should be able to show who can act, on which decision, and how that action changes the outcome.

Evidence: technical paper §12 (timing), §13 (interpretation comparisons), §14 (target dispositions, proof audit and positive-result shortfall), §15 (reproduction). Formal checkpoint: 0b938e373dbf538fe0853f6282e4d9502e67ea01, Lean 4.35.0. The machine/equipment story illustrates the existing abstract high-stakes model. The technical paper and Lean proofs are unchanged.